

ARCH TECHNOLOGIES

Sharpening your hidden skills for a brighter future

Computer Vision

Computer vision involves learning techniques to enable computers to interpret and process visual data from images or videos, using tools like OpenCV and frameworks like TensorFlow or PyTorch. You'll learn concepts like image processing, feature detection, object recognition, and deep learning for tasks such as facial recognition or autonomous driving. It equips you to build systems that analyze and understand visual information for applications in robotics, healthcare, and more.

Task 3: Optical Character Recognition (OCR)

Use Python to build a basic OCR system that can extract text from images containing printed or handwritten text. Work with libraries like Tesseract and OpenCV to preprocess the images, apply OCR, and display the extracted text. Test the system on various sample images and see the results. Dataset to use: [IAM Handwritten Forms Dataset on Kaggle.](#)

Task 4: Facial Emotion Recognition

Build a facial emotion recognition system that can identify emotions such as happy, sad, angry, or surprised from images of human faces. Use a labeled dataset of facial expressions, preprocess the images, train a convolutional neural network (CNN), and evaluate the model's accuracy in classifying emotions. Dataset to use: [FER-2013 dataset](#)

Submission Details:

- Make a ZIP or RAR file of the Code or any related things (if any), or share a GitHub repository link. Email it to submissions.archtech@gmail.com before the 27th of this month.
- For any technical queries or challenges faced during task completion, Email your questions to queries.archtech@gmail.com Our team will do our best to assist you.